PRASHRAY NAGAR

919-264-2726 | nagarprashray16@gmail.com | linkedin.com/in/nagarprashray | github.com/nagarprashray | Berkeley, CA, 94704

Digital ASIC / FPGA Design & Verification Engineer with hands-on experience in SystemVerilog/Verilog RTL microarchitecture, UVM-based functional verification, and FPGA emulation (Xilinx Vivado). Proven track record architecting reusable peripheral IP blocks (UART, SPI, I2C, USB), implementing constrained-random verification environments with SVA/coverage metrics, and resolving CDC/timing closure bottlenecks. Published IEEE researcher in energy-efficient VLSI architectures.

EDUCATION

North Carolina State University, Raleigh, NC || Master of Computer Engineering — Aug 2024 – May 2026
Courses: ASIC and Verification Design, Microprocessor Architecture, Parallel Computing — **GPA:3.3/4.0**

Charusat University, Anand, India || Bachelor of Electronics and Communication — Jul 2020 – May 2024
Courses: VLSI Systems and Design, Embedded Systems, Networking, Data Structures, Compiler Design — **GPA:3.5/4.0**

SKILLS

HDLs & Verification: SystemVerilog, Verilog, VHDL, UVM, Cocotb, SVA, Constrained-Random Testing, Mutation Testing
Architecture & Protocols: SoC Prototyping, CDC/RDC, AMBA (AXI4/Lite, AHB, APB), PCIe, USB, I2C/SPI
EDA Tools (Digital & Sign-Off): Synopsys Design Compiler, PrimeTime, SpyGlass, VC Formal, Formality, Genus, QuestaSim, Verilator
Hardware & Mixed-Signal/RF: Cadence Virtuoso, Keysight ADS, Ansys HFSS, MATLAB, Zynq, UltraScale+, Vivado, Quartus
Programming & Environment: C, C++, Python, Bash, TCL, Make, Git, Docker, Linux

WORK EXPERIENCE

FPGA Engineering Intern, Chipchop, California, USA — **Jul 2026 - Present**
- Spearheaded agentic AI workflows and developed an RTL Testbench Mutation Grader to evaluate agent-generated ASIC testbenches; deployed targeted test cases and defined advanced debug heuristics, achieving over 70% fault-detection accuracy across 5 real production-representative IP-block scenarios.
- Executed end-to-end design, validation, and synthesis for an FPGA-based SoC prototype targeting an upcoming ASIC tape-out; structured peripheral IP architectures for strict portability across both FPGA and ASIC flows to significantly accelerate silicon bring-up.
- Authored reusable peripheral IP blocks (UART, SPI, GPIO, and USB) in Verilog and SystemVerilog; successfully integrated each into the central SoC bus fabric and validated complete on-target functionality prior to final ASIC handoff.

System Engineering Intern, Centre for Environment Education (CEE), Ahmedabad, India — **Jul 2023 – Oct 2023**
- Developed embedded hardware interfaces and firmware communication routines across distributed microcontrollers utilizing SPI, I2C, and UART serial protocols.
- Configured real-time OS (RTOS) tasks and low-latency interrupt handling for multi-sensor data acquisition, synchronizing peripheral actuators with sub-100ms response times.
- Engineered autonomous power-efficient controller nodes, integrating embedded edge processing algorithms that reduced system false-alarm rates while operating within strict thermal and power limits.

PROJECTS

I2C Multi-Bus Verification Using Layered UVM Testbench
- Designed a configurable I2C Multi-Bus (IICMB) Master controller in VHDL, targeting SoC integration via Avalon-MM and Wishbone bus interfaces.
- Built a layered UVM testbench in SystemVerilog — agents, sequencers, and a scoreboard — to verify multi-master, multi-bus IICMB behavior.
- Automated regression testing across modular UVM components (agents, drivers, monitors), reaching 92% functional coverage validated through assertion-linked coverage analysis.

Simulating MESI and MOESI Cache Coherence Protocols
- Developed a cycle-accurate shared-memory hardware simulator to analyze and resolve bus contention bottlenecks across symmetric multiprocessing (SMP) systems.
- Modeled snooping-based MESI and MOESI finite state machines in C++, instrumenting invalidation queues, cache-to-cache data transfers, and memory-bus arbitration under parallel execution traces.
- Demonstrated a 13% reduction in memory-bus traffic and lower transaction latency under MOESI by leveraging Owned-state cache-to-cache data sharing.

C++ Branch Predictor Simulator for Execution Trace Analysis
- Designed an architectural branch predictor simulator to evaluate speculative execution performance and reduce pipeline flush penalties in high-performance superscalar cores.
- Implemented and benchmarked Bimodal, Gshare, and Tournament/Hybrid prediction algorithms in C++, optimizing Pattern History Table (PHT) indexing and Branch Target Buffer (BTB) parameter sizing against SPEC benchmarks[cite: 1, 2].
- Reached 90% prediction accuracy, delivering a measurable 1-point improvement over the bimodal baseline and reducing simulated pipeline stalls

PUBLICATION

"Emerging VLSI Technologies for High-performance AI and ML Applications": Presented at 2024 International Conference on Advancements in Smart, Secure and Intelligent Computing (ASSIC) 2024, the paper explores VLSI hardware accelerators and low-power designs to optimize AI/ML performance and energy efficiency.